Habeeb Chishti

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github.com/habibsnippets

Summary

Applied **Computer Vision / Deep Learning Engineer** with production experience shipping **medical imaging** and **autonomous perception** systems. Expert in bridging research to production: optimizing transformer-based detection (RT-DETR, RF-DETR) and 3D LiDAR models for edge deployment on **NVIDIA Jetson**.

Experience

Machine Learning Engineer (Computer Vision), RNT Health Insights (On-Site) June 2026 – Present

- Migrated GI endoscopy pipelines from RT-DETR to **RF-DETR** to achieve a better balance of detection accuracy and real-time throughput, enabling live clinical decision support during procedures.
- Deployed a fully on-device **speech-to-report pipeline** (MedASR + Qwen LLM) on **NVIDIA Jetson AGX Orin** to automate clinical documentation, reducing physician workload and ensuring patient data privacy by eliminating cloud dependency.
- Engineered embedding-based **frame deduplication** pipelines (DINO/ResNet50 + DBSCAN) to eliminate redundant frames in endoscopy videos, accelerating training cycles and improving model generalization on rare anomalies.
- Optimized multimodal AI pipelines (Vision + ASR + LLM) for embedded GPUs to meet strict hospital latency requirements, enabling real-time assistance during procedures.

Deep Learning Engineer, Flux Auto (YC S21) (On-Site) May 2025 – May 2026

- Integrated **3D LiDAR models** (PointNet, PointNet++, Point Transformer) into the autonomous vehicle stack to enable reliable object detection in outdoor environments where camera-only perception failed due to variable lighting.
- Designed a **Progressive Morphological Filter (PMF)** ground removal pipeline to isolate true obstacles from road noise, preventing false positives and ensuring safe autonomous braking decisions.
- Deployed **low-light image enhancement** algorithms to recover visual data in degraded environments, maintaining camera-based perception redundancy when LiDAR data was sparse or noisy.

Research Intern, SMSS Lab – IIT Kanpur (On-Site) Dec 2023 – Feb 2024

- Developed computer vision algorithms on a robotic platform to autonomously detect and read physical gauges in hazardous field environments, eliminating the need for manual human inspection.
- Fine-tuned a custom **YOLO keypoint + object detection** model on a curated field dataset to localize gauge parameters in real-time, enabling automated, continuous monitoring.

Projects

mathguard — RL Alignment for Mathematical LLMs GitHub

- Built an RLHF-style alignment pipeline for math LLMs to prevent adversarial misuse (e.g., generating harmful recipes) while preserving mathematical reasoning, raising harm refusal from **0% to 88%** with only a **3.2% alignment tax** on GSM8K.
- Designed HarmMathBench and trained a DistilBERT reward model to automate safety evaluation, successfully blocking **100% of 96 jailbreak attempts** (persona, many-shot) on a 1.5B-parameter model.

Rasengan-XL — PEFT Fine-Tuning of SDXL GitHub

- Fine-tuned Stable Diffusion XL using **PEFT** (LoRA, QLoRA, DoRA) to enable custom style-transfer on consumer-grade hardware (16GB T4), making high-fidelity generative AI accessible without enterprise GPU costs.
- Engineered a memory-efficient pipeline (4-bit NF4, gradient checkpointing) to fit a 2.6B-parameter model on limited VRAM, maximizing resource utilization without sacrificing output quality.

PDF2Podcast — Document-to-Audio Generative Pipeline GitHub

- Built a document-to-audio pipeline using Llama-3.2-3B to convert dense technical PDFs into engaging two-speaker dialogues, catering to auditory learners and enabling hands-free content consumption.
- Designed a conversational framework where one speaker synthesizes concepts while the other drives engagement via follow-up questions, rendering broadcast-quality audio via Kokoro KPipeline TTS.

Education

National Institute of Technology, Srinagar — B.Tech 2021 – 2025

- GPA: 8.311 / 10.0

Technical Skills

Languages: Python, C++, C

Deep Learning & CV: PyTorch, Hugging Face, RT-DETR, RF-DETR, YOLO, ResNet, ViT, DINO, OpenCV, timm

3D / Point Cloud: PointNet, PointNet++, Point Transformer, LiDAR processing, PCL

LLMs & GenAI: Llama, Qwen, LangChain, RLHF, GRPO, LoRA / QLoRA / DoRA, Stable Diffusion XL

Edge & MLOps: NVIDIA Jetson AGX Orin, CUDA, TensorRT, ONNX, Docker, W&B, Linux